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Inventor(s)

Schaeffer; Benjamin et al.

OPERATING INTELLIGENT COMPLETION VALVES IN A HYDROCARBON WELL USING MACHINE LEARNING

Abstract

A system for operating intelligent completion valves (ICVs) used in a hydrocarbon well operation is disclosed. The ICVs can be positionable downhole in a wellbore. A trained machine-learning model can be generated by training a machine-learning model on training data comprising acoustic signal data generated by a pressurized injection fluid flowing through the ICVs at various flow rates and different ICV positions. When new acoustic sensing system sensor data associated with an ICV of multiple ICVs in the wellbore is applied to the trained machine-learning model, the trained machine-learning model can generate a result determining that the amplitude spike is attributable to a change in the position of the intelligent completion valve. The system may also determine a magnitude of the change in the position of the intelligent completion valve. The result may be output and used to control the ICV.

Inventors: Schaeffer; Benjamin (Denver, CO), Jaaskelainen; Mikko K. (Houston, TX), Jugdaw; Sanjay (Spring, TX)

Applicant: Halliburton Energy Services, Inc. (Houston, TX)

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates generally to wellbore operations, and more particularly (although not necessarily exclusively) to operating intelligent completion valves using machine learning.

BACKGROUND

[0002] Hydrocarbon well operators may design and construct wells having intelligent completions, which can allow the well operator to control downhole processes from the surface. For example, an intelligent completion may include control valves, sensors, or other devices via which downhole production or injection processes can be controlled and monitored. An operator at the well surface may adjust the positions of downhole control valves via hydraulic fluid delivered using hydraulic control lines. But the operator may be unaware of previous control valve setting changes, and the characteristics of the hydraulic fluid can change based on temperature and other factors. It also may be difficult for an operator located at a well surface to determine if a given control valve has moved or reached an intended setting position. As a result, a given control valve adjustment made from the well surface may not produce the intended or expected result.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. 1 is a schematic diagram of a hydrocarbon well system according to one example of the present disclosure.

[0004] FIG. 2 is a block diagram depicting one example of a computing device for operating downhole intelligent completion valves, according to an example of the present disclosure.

[0005] FIG. 3 is a block diagram of an application for training a machine-learning model to generate an output that can be used to operate downhole intelligent completion valves, according to an example of the present disclosure.

[0006] FIG. 4 is a block diagram of a machine-learning-based system including the computing device of FIG. 2. for operating downhole intelligent completion valves, according to an example of the present disclosure.

[0007] FIG. 5 is a flow chart representing one example of a method for operating downhole intelligent completion valves according to an example of the present disclosure.

DETAILED DESCRIPTION

[0008] Certain aspects and examples of the present disclosure relate to a machine-learning-based system for operating intelligent completion valves (ICVs) that are usable downhole in a hydrocarbon well. The ICVs can be used, for example, to control inflow or injection relative to a given well. In one example, the ICVs can be located in a completed injection well that is drilled in a formation and into which is directed a pressurized injection fluid. The pressurized injection fluid can be emitted from the injection well and injected into the formation through the ICVs and corresponding perforations in the injection well. The flow rate of pressurized injection fluid into the formation may be controlled, at least in part, by the ICVs, which can allow target flow rates through the formation to be maintained.

[0009] The flow of pressurized injection fluid through the ICVs produces resulting acoustic signals. An acoustic sensing system may be used to record the acoustic signals produced by the ICVs. The acoustic signals may include amplitude spikes and other anomalies that may be indicative of ICV position changes such as choke setting changes, and the correlation between these acoustic signal anomalies and ICV functions can be recognized and utilized by the system. The flow rate of the

pressurized injection fluid within the injection well can also be measured in the area of each ICV and recorded concurrently with recording by the acoustic sensing system of the acoustic signals generated by the ICVs. The recording of the flow rates and the acoustic signals may occur at different ICV positions to correlate the pressurized injection fluid flow rate data and ICV acoustic signal data with the ICV position data. Likewise, the recording of acoustic signals may occur at different pressurized injection surface fluid flow rates to correlate acoustic signal data with pressurized injection fluid flow rate data. The recording of the flow rates and the acoustic signals may occur at different pressurized injection fluid pressures. In some examples, the flow rate of the pressurized injection fluid within the injection well can be measured using an injection logging tool.

[0010] In some examples, acoustic signal data and pressurized injection flow rate data may be obtained from a well other than a well of current interest. In other examples, the training data may be data obtained directly from the well of current interest, such as by operating an injection logging tool in the well of current interest to record pressurized injection fluid flow rates while simultaneously recording ICV acoustic signal data using an acoustic sensing system.

[0011] Recorded acoustic signal data and corresponding recorded pressurized injection fluid flow rate data can serve as training data for training a machine-learning model. More specifically, a trained machine-learning model can be generated by training the machine-learning model on training data that includes the recorded acoustic signal data generated by the flow of pressurized injection fluid through the ICVs and the corresponding measured flow rates of the pressurized injection fluid, at different ICV positions and possibly different pressurized injection fluid pressures. Other data, such as various engineered feature data, may also be part of the training data in some examples.

[0012] New sensor data received by the distributed acoustic sensing system in response to an amplitude spike in a new acoustic signal associated with an ICV can thereafter be applied to the trained machine-learning model. Based on the input of the new sensor data, the trained machine-learning model can generate a result determining that the amplitude spike is attributable to a change in the position (i.e., movement) of the ICV.

[0013] In some examples, the system can also determine the extent (magnitude) of an ICV position change. For example, and as explained in more detail below, the system may associate the observed intensity (frequency) of an acoustic emission generated by passage of the pressurized injection fluid through an ICV with a known flow rate of the pressurized fluid at the time of the observation. This process can be repeated for the various ICVs in a wellbore, at different ICV positions and different pressurized fluid flow rates, to create a collection of acoustic signatures for each ICV. With the flow rate of the pressurized fluid known, these acoustic signatures can thereafter be used to infer an ICV position based on the measured intensity of an acoustic emission of the ICV. A difference between the acoustic signature of an ICV prior to a position change and the acoustic signature of the ICV after a position change can be used to infer a magnitude of the position change (movement) of the ICV.

[0014] The system may use the result generated by the trained machine-learning model to control the associated ICVs in a well. For example, the system may further adjust a position of an ICV based on confirmation of movement of the ICV. In other examples, the system may provide the result generated by the trained machine-learning model to another control system, which can use the information conveyed by the result to control the associated ICVs in a similar manner.

[0015] Operating ICVs can be used to increase efficiency of the wellbore operation. For example, an outflow rate of pressurized injection fluid from an injection well into a formation that is too high may result in water breakthrough, where hydrocarbons in the formation are displaced by water that can migrate to the producing well(s) and reduce the amount of hydrocarbons extracted from the formation. Injecting too much injection fluid into the formation may also cause more severe issues such as a loss of caprock integrity, which can lead to problems with hydrocarbon envelope

containment. Operating ICVs of a hydrocarbon well operation may also allow target flow rates to be more easily achieved and maintained, which can allow the well field to be more accurately managed and can improve production and sweep efficiency.

[0016] Illustrative examples follow and are given to introduce the reader to the general subject matter discussed herein rather than to limit the scope of the disclosed concepts. The following sections describe various additional features and examples with reference to the drawings in which like numerals indicate like elements, and directional descriptions are used to describe the illustrative aspects, but, like the illustrative aspects, should not be used to limit the present disclosure.

[0017] One example of a hydrocarbon well field **100** is represented in the diagram of FIG. **1**. As shown, the hydrocarbon well field includes a completed injection well **102** and at least one completed offset producing well **150** located in a reservoir. The injection well **102** can include a wellbore **104** that is drilled in a subterranean formation **106** within the reservoir, but may alternatively be drilled in a sub-oceanic formation in other examples. In this example, the injection well **102** includes only a vertical wellbore **104**. In other examples, the injection well **102** may also have a horizontal wellbore portion. The injection well **102** may include a wellbore casing **108**, which may be cemented into the wellbore **104** by introducing cement **110** into an annular space between the wellbore **104** and wellbore casing **108**.

[0018] The injection well **102** may include multiple sets of perforation clusters **112** that are disposed at intervals along the wellbore **104** and pass through the wellbore casing **108** and the cement **110**. In this example, a set of perforation clusters **112** coincides with each of multiple treatment stages of the injection well **102**, with each treatment stage being associated with a different zone of the formation **106**. The various treatment stages of the injection well **102** may be separated using plugs or other suitable elements to isolate stages, and each treatment stage may have more than one set of perforation clusters.

[0019] In the case of the completed injection well **102**, the perforation clusters **112** allow pressurized injection fluid **114** to be emitted from the wellbore **104** and injected into the formation **106** as indicated. In this example, the injection fluid is obtained from an injection fluid source **116** (e.g., a tank) and pressurized using one or more pumps **118** before being delivered into the wellbore casing **108** of the wellbore **104**. The pump(s) **118** may be a fixed position pump, one or more mobile pumps such as pump trucks, etc. As can be observed in FIG. **1**, the pressurized injection fluid **114** will flow forcibly outward through the perforation clusters **112** and into the formation **106**. The flow of pressurized injection fluid **114** into the formation **106** can encourage hydrocarbons located in the formation **106** between the injection well **102** and the producing well **150** to migrate toward the producing well **150** for extraction, as indicated by the arrows **120**. In some examples, the injection well **102** may be a water injection well.

[0020] The outflow of the pressurized injection fluid **114** from the injection well **102** into the formation **106** can be controlled by a plurality of intelligent completion valves (ICVs) **122** installed downhole in the injection well **102**. The ICVs **122** may be part of an intelligent completion, which may also include sensors or other devices that can facilitate monitoring and control over the injection well **102**. In at least some examples, a position (i.e., open, closed, partially open) of the ICVs **122** can be adjusted from the well surface **124**. For example, the position of an ICV **122** may be adjusted via hydraulic fluid delivered from the well surface **124** through a hydraulic supply line. A position adjustment of an ICV **122** may be initiated by an operator, a computing device **200**, or a separate ICV (process) control system **250** located at the well surface **124**.

[0021] In the example depicted in FIG. **1**, the ICVs **122** in the injection well **102** may be part of a tubing-conveyed completion that is located in the wellbore **104**. The tubing-conveyed completion may include a tubing string **126**. Multiple isolation elements **128** may be coupled to the tubing string **126** along the length thereof, and may be located to straddle the various perforation clusters **112** as shown. The isolation elements **128** can be used to isolate different zones and corresponding

perforation clusters **112** within the wellbore **104**. In this regard, the isolation elements may be deployed in one or more pairs, where at least one first isolation element is positioned on the uphole side and at least one cooperating isolation element is positioned on the downhole side, of each set of the perforation clusters **112** in the wellbore **104**. For example, as shown in FIG. **1**, the various pairs of isolation elements **128** may be used to isolate four separate zones, denoted as Zone **1** through Zone **4**, where each isolation zone encompasses a perforation cluster **112**. The isolation elements **128** may be individually controllable or may be controllable in cooperating pairs, to isolate an individual zone of the wellbore **104** around a given set of the perforation clusters **112** or to simultaneously isolate multiple zones of the wellbore **104** around several sets of the perforation clusters **112**. In such an example, one or more ICVs **122** may be located on the tubing string **126** between each cooperating pair of isolation elements **128**. The ICVs **122** may, but are not required to be, positioned to be substantially aligned with the perforation clusters **112** bounded by the isolation elements **128**, to facilitate passing of the pressurized injection fluid **114** emitted by the ICVs **122** through the perforation clusters **112** and into the formation **106**.

[0022] By employing a tubing-conveyed completion having multiple isolation elements **128** as described, the flow of pressurized fluid may be allocated on a per ICV basis, a per isolation zone basis, or a per cluster basis. In some examples, the isolation elements **128** may be inflatable packers, the inflation and deflation of which may be controllable from the well surface **124**. In some examples, the isolation elements **128** may be swell packers that can expand once in contact with wellbore fluids such as brine, water, or hydrocarbons. While the example of FIG. **1** depicts the injection well **102** as having four isolation zones with one perforation cluster **112** per zone shown, it should be understood that a different number of isolation zones and a different number of perforation clusters **112**, including a different number of perforation clusters per isolation zone, may be present in other well examples.

[0023] As with the injection well **102**, the producing well **150** may include a wellbore **152** that is drilled in the subterranean formation **106**. In this example, the producing well **150** includes only a vertical wellbore **152**. In other examples, the producing well **150** may also have a horizontal wellbore portion. The producing well **150** may include a wellbore casing **154** for transporting hydrocarbons **156** to the well surface **124**. The wellbore casing **154** of the producing well **150** may also have cement **158** located between the wellbore casing **154** and the wellbore **152**. The completion of the producing well **150** may include a wellhead **162** (e.g., Christmas tree) or another well completion apparatus such as a pump, a derrick, etc.

[0024] The producing well **150** may also include multiple sets of perforation clusters **160** that are disposed at intervals along the wellbore **152** and pass through the wellbore casing **154** and the cement **158**. The sets of perforation clusters **160** may again coincide with each of the multiple treatment stages of the injection well **102** and with the different zones of the formation **106**. In the case of the producing well **150**, the perforation clusters **160** can allow the hydrocarbons from the formation **106** to enter the producing well **150** and to be transported to the well surface **124**.

[0025] The outward flow of the pressurized injection fluid **114** through the perforation clusters **112** and the ICVs **122** of the injection well **102** creates acoustic emissions at the ICVs, the properties of which can be captured as acoustic signals. The particular sound and intensity (amplitude) of a given acoustic emission is related to the position of the ICV **122** through which the pressurized injection fluid **114** passes. For example, the amplitude of an acoustic signal generated by the outward flow of the pressurized injection fluid **114** through an ICV **122** and an associated set of perforation clusters **112** may increase as the degree to which the ICV **122** is opened increases.

[0026] The acoustic signal associated with the flow of the pressurized injection fluid **114** through the perforation clusters **112** and the ICVs **122** of the injection well **102** can be recorded for each ICV **122** using an acoustic sensing system **130**. In some examples, the acoustic sensing system **130** can collect DAS data, DTS data, downhole pressure gauge data, surface pressure gauge data, or surface injection flow data. The recorded differential phase (acoustic) DAS data may be a 2-

dimensional measurement of strain rate with respect to measured depth and time.

[0027] In the example of FIG. 1, the acoustic sensing system **130** is a distributed acoustic sensing (DAS) system. The DAS system can include an optical fiber cable **132** that is located in the wellbore **104** of the injection well **102**. In some examples, the optical fiber cable **132** can be clamped to or cemented to the outside of the wellbore casing **108**. In other examples, the optical fiber cable **132** can be deployed into the casing **108** of the injection well **102** along with a wireline, a slickline, or a permanent cable, may be deployed along with and clamped or otherwise affixed to the tubing string **126** as shown, or may be deployed by another suitable technique. The optical fiber cable **132** may also be deployed into the injection well **102** by gravity, pumping, or by pushing or otherwise inserting the optical fiber cable **132** into the injection well **102**. The optical fiber cable **132** may also be self-propelled using mechanical or chemical propulsion. The optical fiber cable **132** can be various types of optical fiber cables. For example, the optical fiber cable **132** may be a lower cost disposable optical fiber cable.

[0028] The optical fiber cable **132** extends, in this example, from a wellhead exit of the wellbore casing **108**, along the surface **124** of the formation **106**. A surface optical fiber cable **134** can connect the optical fiber cable **132** attached to the completion tubing **126** with an optoelectronic interrogator of the DAS system. The optical fiber cable **132** and the optoelectronic interrogator may form at least a primary portion of the DAS system. The DAS system may utilize various implementations of, for example, Rayleigh or Brillouin scattering, and may be interferometric in nature. Employed sensing technologies may include, for example, homodyne, heterodyne, Michelson, Mach-Zender, Fabry-Perot, phase based, intensity based, coherence based, static (absolute) or dynamic (relative) sensing principles.

[0029] The acoustic sensing system **130** may be communicatively coupled to a computing device **200**. The computing device **200** can receive acoustic signal data from the acoustic sensing system **130** and may use the acoustic signal data within the process of controlling the ICVs **122**. In some examples, the computing device **200** may also use the acoustic signal data to train a machine-learning model.

[0030] As illustrated in FIG. 1, the intelligent completion may also include an ICV control system including an ICV control device **250**. The ICV control device may be a computing device, and may be communicatively coupled to the downhole ICVs **122** by one or more ICV control cables **136**. The one or more ICV control cables may be one or more wires, one or more optical fiber cables, one or more hydraulic control lines, or another mechanism via which at least control signals from the ICV control device **250** may be transmitted to the ICVs **122**. The ICV control cable(s) **136** may be deployed into the casing **108** of the wellbore **104** in any manner described above relative to the optical fiber cable **132** of the acoustic sensing system **130**. In this example, the ICV control cable(s) **136** is affixed, such as by clamping, to the outside of the tubing string **126**. Although the ICV control cable(s) **136** and the optical fiber cable **132** of the acoustic sensing system **130** are shown to be on substantially opposite sides of the tubing string **126** for clarity in FIG. 1, it is possible that the ICV control cable(s) **136** and the optical fiber cable **132** of the acoustic sensing system **130** may be co-located in other examples, such as in a flat pack that is clamped to the tubing string **126**.

[0031] As further illustrated in FIG. 1, the ICV control cable(s) **136** may extend from the wellhead exit of the wellbore casing **108**, and along the surface **124** of the formation **106**, in a similar manner to the optical fiber cable **132** of the acoustic sensing system **130**. A surface ICV control cable **138** can connect the optical ICV control cable **136** attached to the ICV control device **250**. The ICV control device **250** may also be communicatively coupled to the computing device **200**, such that the ICV control device **250** may receive information from the computing device **200** that may be used by the ICV control device **250** to control a position of one or more of the ICVs **122**. In this manner, the acoustic sensing system **130**, the ICV control device **250**, and the computing device **200** can cooperatively form a closed loop control system. As described in more detail below, the closed loop control system may include exception handling functionality.

[0032] When the acoustic detection system **126** is a DAS system, for example, acoustic signal data may be acquired at a data rate of 1 kHz to 100 kHz depending on the optical fiber length, and more commonly between 5 kHz to 20 kHz. The data rate of a DAS system is normally significantly faster than any possible movement of an ICV and associated acoustic signals generated by the ICV movement, which can allow acoustic signals to be stacked and processed. ICVs may, in some instances, be moved in non-intuitive sequences between control points/valve positions, which may yield additional information that can be used to confirm ICV positions and system operation. For example, acoustic signal data generated as a result of such non-intuitive movement sequences can be used to tune control algorithms or improve ICV position prediction.

[0033] The flow rate of the pressurized injection fluid **114** flowing outward through the individual ICVs **122** of the injection well **102** of FIG. **1** can be measured under known conditions (e.g., surface temperature and pressure, injection fluid temperature, etc.). For example, flow rate data associated with each of the ICVs **122** may be measured by an injection logging tool that is temporarily deployable downhole in the injection well **102** and configured to measure the flow rate of the pressurized injection fluid at various locations within the injection well **102**. The injection logging tool can be used to determine the known flow rates of the pressurized injection fluid **114** through the ICVs **122** by measuring the flow rates of the pressurized injection fluid within the completed injection well **102** at locations near each of the ICVs **122**. The location (e.g., depth) of each ICV **122** is known, whether from injection well construction data, as a result of an inspection/logging operation, or otherwise. As such, a measured pressurized injection fluid flow rate can be associated with a given ICV **122** based on the location of the injection logging tool within the injection well **102** at the time of measurement.

[0034] The acoustic sensing system **130** may be operated to record ICV acoustic signal data generated by the pressurized injection fluid flowing through the ICVs at the same time the injection logging tool is used to record pressurized injection fluid flow rates. The location of each ICV **122** within the injection well **102** is identifiable by the acoustic sensing system, thus, received acoustic signal data can be properly associated with the correct ICV **122**. Both acoustic signal data and pressurized injection fluid flow rate data for each individual ICV **122** can be measured and stored. For example, when operating the injection logging tool, the ICVs **122** may be controlled such that the pressurized injection fluid **114** flows through only one of the plurality of ICVs **122** at a time. Additionally, the positions of the ICV **122** through which the pressurized injection fluid flows may be changed in a step-wise fashion to provide a collection of acoustic signal data and pressurized injection fluid flow rate data for each ICV **122** at each of the various possible positions thereof.

[0035] The acoustic signal data recorded by the acoustic sensing system **130** can be translated to the flow rates of the pressurized injection fluid **114** flowing through the plurality of intelligent completion valves as measured by the injection logging tool. In other words, a correspondence between an acoustic signal and a pressurized injection fluid flow rate through a given ICV **122** can be established. Selected data may be stored locally, where characteristic signatures for different ICV positions and ICV position (e.g., choke setting) changes may be tracked over time. Similarly, ICV setpoint data may be used as input for injection fluid property compensation once a suitable dataset has been achieved.

[0036] The ICV acoustic signal data and the pressurized injection fluid flow rate data may be obtained from a well of current interest (e.g., the injection well **102** in which the ICVs **122** to be controlled are located), as described above. For example, an injection logging tool may be operated in the injection well **102** to record pressurized injection fluid flow rates while simultaneously recording ICV acoustic signal data using the acoustic sensing system **130**. In other examples, acoustic signal data and pressurized injection flow rate data may be obtained from a well other than a well of current interest—i.e., from a generic well. The generic well may be selected to have similar features to the injection well **102**. For example, the generic well may be located in similar environmental conditions, may be located in a similar formation, may have a similar set of ICVs,

may be injected with a similar pressurized injection fluid at similar pressures, etc. In some examples, the acoustic signal data and the pressurized injection flow rate data obtained from the generic well may thus serve as baseline training data for modeling ICV operation in a well of current interest.

[0037] To measure the effects of a change in position of an ICV, a relative amplitude technique that evaluates the root-mean-square (RMS) of ICV acoustic signal amplitudes before and after a detected time of position change may be used. For example, taking the median filtered data and summing in time (without moveout correction) provides for a ratio of the RMS amplitudes of the ICV acoustic signal before and after a position change. This can indicate whether changing the ICV position has had any effect and can also validate whether a detected ICV position change was a false positive. According to some examples, if the ratio of the RMS amplitudes meets a specified threshold (e.g., opening ICV valve $>x$, closing ICV valve $<y$) then the difference in the RMS amplitude between the starting position of the ICV valve and all subsequent positions can be taken, and a percentage change can be developed.

[0038] When a change in position of an ICV **122** is initiated from the well surface **124**, the initial responsive movement of the ICV **122** will normally cause a spike in the amplitude of the acoustic signal that is generated by the ICV **122** and is detectable by the acoustic sensing system. More specifically, an opening or closing movement of the ICV **122** can generate a measurable acoustic response, which propagates uphole and downhole from the ICV position in measured depth. It is this measurable acoustic response resulting from movement of an ICV **122** that is referred to herein generally as a spike.

[0039] The frequency content of the acoustic signal is typically largely broadband in nature, but a low pass filter can be applied at a frequency of approximately 500 Hz, for example, to capture the signal for subsequent processing. Data associated with the captured and filtered acoustic response signal can be truncated about each ICV **122** in measured depth between an uppermost and a lowermost bounding ICV **122** of the ICVs **122** in the wellbore **104**. Following truncation of the acoustic response signal data in measured depth, a traveltime moveout correction can be applied about the position of the moved ICV **122** to align the resulting acoustic signal in time. A variety of attributes can be extracted from the aligned acoustic signal to capture the impulsive nature of the acoustic signal. One such attribute may be the short term average to long term average amplitude ratio (STA/LTA) of the acoustic signal. The STA/LTA ratio may be computed and the peak amplitude value of the acoustic signal can be measured by, for example, summing across the time aligned and truncated acoustic signal data while employing a coherency weighting technique (semblance). The peak amplitude value of the acoustic signal can be compared to a predetermined threshold value to determine whether the acoustic response signature matches that of an ICV opening or closing event or a (coherent or incoherent) noise event.

[0040] The locations of the ICVs of a given hydrocarbon well are known in space (e.g., measured depth) and thus acoustic signal data associated with the ICVs can be truncated about the ICV positions within a defined length of time. For example, the acoustic signal data associated with the ICVs can be truncated by multiplying the acoustic signal data samples in time by the number of ICVs in the well and by the number of truncated recording channels selected or present.

[0041] An amplitude spike in an ICV acoustic signal may be followed by a general increase or a general decrease in the amplitude of the acoustic signal over time. In some examples, an amplitude spike in an ICV acoustic signal followed by a general decrease in amplitude of the acoustic signal over time may indicate an ICV closing event, while an amplitude spike in the acoustic signal followed by a general increase in amplitude of acoustic signal over time may indicate an ICV opening event. When such a general increase or a general decrease in the amplitude of an acoustic signal over time is detected, a rate of change in the amplitude of the acoustic signal may be monitored and it can be determined when the rate of change becomes stable. If the rate of change in the amplitude of the acoustic signal becomes stable within a predefined length of time following

the amplitude spike in the acoustic signal, the flow rate of the pressurized injection fluid at the time the rate of change in the amplitude of the acoustic signal became stable may be assigned as the flow rate of the pressurized injection fluid through the ICV at the new (current) position of the ICV. [0042] When processing a frame of truncated recorded ICV acoustic signal data, several filtering steps can be applied to maximize the signal-to-noise ratio and preserve the acoustic energy that corresponds to the flow of pressurized injection fluid through the ICVs. For example, a 2D median filter with a pre-defined kernel size may first be applied. In some examples, the absolute value of the ICV acoustic signal data may then be taken, followed by application of a narrow band bandpass filter. To generate engineered features, a beamforming technique can then be applied by moveout correcting the data about a midpoint ICV channel utilizing the velocity of sound through fluid (water) within a pipe (flattening the data in time and computing a stack amplitude-semblance metric). Although a beamforming technique is described above with respect to the velocity of sound through water, it should be understood that the beamforming process is also applicable to other fluids such as, for example, liquid carbon dioxide (CO₂), liquid natural gas (LNG), or other hydrocarbons. Once the acoustic signal data is flattened, various other anomaly detection feature engineering methods can be used. For example, the absolute value of the data may be taken and then summed across all truncated channels before applying a characteristic function (e.g., short time over long time average ratio). Stacking the amplitude signal data can help to accentuate the acoustic signal of the ICV across the multiple channels while canceling out noise within single channels.

[0043] Acoustic signal data associated with the ICVs **122** and recorded by the acoustic sensing system **130** can be combined with recorded pressurized injection fluid flow rate data measured by the injection logging tool to build training data that is usable to train a machine-learning model. The training data may also include other data, such as various engineered features data. Engineered features that may be included in the training data may include, for example, moveout-corrected stack semblance, ratio of short time average to long time average, ratio of the root-mean-square amplitude of the acoustic signal before an amplitude spike to the root-mean-square amplitude of the acoustic signal after an amplitude spike, and combinations thereof. In some examples, the training data may additionally include surface conditions such as temperature or pressure, injection fluid temperature or flow rate, etc. Some of the recorded data may also be used in real time to update allocated pressurized injection fluid flow values, as the surface flow rates and pressures may fluctuate.

[0044] FIG. 2 is a block diagram depicting one example of a computing device **200** that is usable, in conjunction with other components, to operate an ICV. While FIG. 2 shows the computing device **200** as including certain components, other examples may involve more, fewer, or different components than are shown in FIG. 2. The computing device **200** may be communicatively coupled to an ICV control device, such as the ICV control device **250** of FIG. 1, that may be configured to electrically or hydraulically actuate ICVs. The computing device **200** is shown to include a processor **202** that may be communicatively coupled to a memory **204** by a bus **206**. The processor **202** can include one processor or multiple processors. Non-limiting examples of the processor **202** include a Field-Programmable Gate Array (FPGA), an application specific integrated circuit (ASIC), a microprocessor, or any combination of these. Instructions **208** may be stored in the memory **204**. Output generated by the processor **202** can be presented via various mediums, including on a display device **210** that is communicatively coupled to the processor **202** by the bus **206**.

[0045] The instructions **208** are executable by the processor **202** for causing the processor **202** to perform various operations. In some examples, the instructions **208** can include processor specific instructions generated by a compiler or an interpreter from code written in any suitable computer-programming language, such as C, C++, C #, Java, or Python. Through the instructions **208**, the processor **202** may operate as described above to perform the various operations of the computing

device **200** related to ICV operation.

[0046] The memory **204** can include one memory device or multiple memory devices. The memory **204** can be non-volatile and may include any type of memory device that retains stored information when powered off. Non-limiting examples of the memory **204** include electrically erasable and programmable read-only memory (EEPROM), flash memory, or any other type of non-volatile memory. At least some of the memory device includes a non-transitory computer-readable medium from which the processor **202** can read the instructions **208**. A non-transitory computer-readable medium can include electronic, optical, magnetic, or other storage devices capable of providing the processor **202** the instructions **208** or other program code. Non-limiting examples of a non-transitory computer-readable medium include magnetic disk(s), memory chip(s), ROM, random-access memory (RAM), an ASIC, a configured processor, optical storage, or any other medium from which a computer processor can read the instructions **208**.

[0047] FIG. **3** is block diagram illustrating one example of a model-training application **300** that can be executed by the computing device **200**, or by a model training subsystem or another computing device that is separate from the computing device **200**, to train a machine-learning model **302**. Training the machine-learning model **302** using the model-training application **300** can transform the machine-learning model **302** from an untrained state to a trained state—i.e., to a trained machine-learning model, such as the trained machine-learning model **402** of FIG. **4**.

[0048] Training the machine-learning model **302** can include accessing training data that may be compiled or otherwise presented in a training dataset **304**. The training data may include at least ICV acoustic signal data **306** that has been recorded by the acoustic sensing system **130** and is associated with the ICVs **122** in the injection well **102**, as well as recorded pressurized injection fluid flow rate data **308** measured by an injection logging tool while deployed downhole in the injection well **102**. The training dataset **304** may also include command signal data **312** associated with command signals sent from the computing device **200** to the ICV actuation system. Other training data such as various engineered features training data **310** described above may also be a part of the training dataset **304**. In some examples, the training dataset **304** can be stored in a data repository that can be accessed by the model-training application **300**.

[0049] Although not illustrated in FIG. **3**, in some examples, the various training data **306**, **308**, **310** within the training dataset **304** can be divided into several parts. For example, the training data **306**, **308**, **310** may be divided into one or more of a training dataset, a validation dataset, and a testing dataset. In any case, the locations of the ICVs **122** and the times of ICV opening and closing and other information relative to the training data **306**, **308**, **310** within the training dataset **304** is known, thus the training data **306**, **308**, **310** can be annotated and prepared for model training. A variety of anomaly detection models can be employed to determine the presence of an ICV position change by analyzing the ICV acoustic signal data **306**, including for example, autoencoder and random forest models.

[0050] The machine-learning model **302** can be trained by causing the model-training application **300** to provide the training data **306**, **308**, **310** as input to the machine-learning model **302**. In some examples, the model-training application **300** may be configured to provide the training data **306**, **308**, **310** from the training dataset **304** to the machine-learning model **302** in an iterative manner. This can help the machine-learning model **302** to identify trends or relationships in the training data **306**, **308**, **310** and to find a set of model parameters that minimize a loss function or another function. Parameters or hyperparameters may also be selected to find the model parameters that optimize the fit between prediction and actual outputs. Various other fitting, estimation, or other model-training optimization techniques can be used to ensure that, upon evaluation, the predictive output of the machine-learning model **302** is accurate given the input data.

[0051] The machine-learning model **302** may be trained in a supervised manner, an unsupervised manner, or a semi-supervised manner. In one example, some or all of the training data **306**, **308**, **310** in the training dataset **304** can be labeled and the machine-learning model **302** can be trained

using a supervised learning technique.

[0052] Execution of the model-training application **300** of FIG. **3** on the training data **306**, **308**, **310** in the training dataset **304** can output a trained machine-learning model **402** (see FIG. **4**) with an optimized set of parameters and hyperparameters for use in machine-learning inference. In the example of FIG. **4**, the trained machine-learning model **402** can be stored on the computing device **200**. Using the trained machine-learning model **402**, the computing device **200** can be used to monitor a well and to generate an output when new unlabeled input data **404** in the form of ICV acoustic signal data **406** is applied to the trained machine-learning model **402**.

[0053] The trained machine-learning model **402** may be configured to generate various types of results. For example, when the new acoustic signal data **406** associated with a given ICV is applied to the trained machine-learning model **402**, the trained machine-learning model **402** can generate a result determining that an amplitude spike in an acoustic signal within the acoustic signal data is attributable to a change in the position of the intelligent completion valve—i.e., determining that an ICV position change occurred **410**. The result generated by the trained machine-learning model **402** can be used by the computing system **200**, at least in part, to control the position of one or more of the ICVs.

[0054] In some examples, the system can also determine the magnitude of an ICV position change. For example, the system may associate a number of acoustic signatures with each ICV in a wellbore. Each acoustic signature may be a combination of an observed frequency of an acoustic emission generated by passage of the pressurized injection fluid through an ICV at a known pressurized fluid flow rate and a known ICV position. Varying the pressurized fluid flow rate, the ICV position, or both, can result in a different acoustic signature. A multitude of acoustic signatures for each ICV corresponding to different pressurized fluid flow rates and ICV positions may be stored for each ICV, and may be included in the training dataset **304** used to train the machine-learning model **302**. The trained machine-learning model **402** may thus generate and analyze ICV acoustic signatures when provided with new input data. To determine a magnitude of an ICV position change, a measured intensity of the acoustic emissions of the ICV at an initial ICV position may first be compared to an acoustic signature of the ICV according to the trained machine-learning model **402** at like conditions (pressurized fluid flow rate, etc.) to infer a current ICV position. A measured intensity of the acoustic emissions of the ICV at the modified position may then be compared to an acoustic signature of the ICV according to the trained machine-learning model **402** at like conditions to infer the new ICV position. The difference between the inferred initial ICV position and the inferred new ICV position can be used to determine a magnitude of the ICV position change **412**. The number of ICV position changes can also be inferred in this manner, any may provide real-time feedback that can allow for optimal ICV position changes to be selected.

[0055] In some examples, the determination of the ICV position change **410** included in the result generated by the trained machine-learning model **402** may be output to an operator or to a separate ICV control system **250**. A determined magnitude of the ICV position change **412** may be output in a same or a similar manner. The ICV control system **250** receiving such information may be, according to some examples, a supervisory computer system for data processing, visualization, or command and control of setpoints related to the production of hydrocarbons from a well or well system with which the ICVs associated with the notification are a part. When received by an ICV control system **250**, the result information output by the trained machine-learning model **402** may be displayed together with surface flow rates, surface pressure, flow allocation per ICV, or other information related to hydrocarbon well operations.

[0056] In some examples, the computing device **200** or the ICV control system **250** may include an automated exception handling routine whereby the computing device **200** or the ICV control system **250** can conditionally take an action. For example, the ICV control system **250** may, after receiving the result generated by the trained machine-learning model **402**, set a target flow

allocation of the pressurized fluid per ICV, minimum/maximum flow rates per ICV, allowable deviations from the target flow allocation before an exception handling request is triggered, or may initially set a starting point of ICV positions to generate the target flow allocation for a given well. [0057] In some examples, the output **408** of the trained machine-learning model **402** may be usable to control a position of an ICV. For example, the output **408** of the trained machine-learning model **402** and subsequent amplitude tracking by an acoustic sensing system may allow the computing device **200** (or the ICV control system **250**) to make fully automated ICV position changes, which can obviate the need for an onsite operator and associated manual ICV position adjustments. Consistent ICV position changes can be better ensured through such automated control and, resultantly, actual flow rates through a formation can be more easily and accurately matched to and maintained at target flow rates.

[0058] FIG. **5** is a flow chart **500** representing a method for operating an ICV located downhole in a completed injection well, according to one example of the present disclosure. As represented in FIG. **5**, at block **502**, a processor of a computing device can receive, from an acoustic sensing system, sensor data associated with an ICV of a plurality of ICVs positionable downhole in a wellbore in a formation. The sensor data may be generated in response to an amplitude spike in an acoustic signal associated with one of the ICVs. In some examples, the completed injection well may be associated with at least one completed offset producing well. In such an example, the injection well may emit pressurized injection fluid into the formation to encourage hydrocarbons located therein to move to the producing well for extraction. In some examples, the pressurized injection fluid can be water.

[0059] At block **504**, the processor can apply the sensor data to a trained machine-learning model. The trained machine-learning model may have been previously trained using training data comprising acoustic signal data generated by a pressurized injection fluid flowing outward through the plurality of ICVs at various flow rates and at different ICV position settings. The acoustic signal data may be recorded by the acoustic sensing system (e.g., a DAS system). In some examples, the training data may be built by operating an injection logging tool downhole in the wellbore to determine the known flow rates of the pressurized injection fluid through the plurality of ICVs while simultaneously operating the acoustic sensing system to record associated acoustic signal data.

[0060] At block **506**, the trained machine-learning model may generate a result determining that the amplitude spike in the acoustic signal data is attributable to a change in the position of the ICV. The magnitude of the change in the position of the ICV may also be determined as described above, and may be indicative of, or may be used to determine a new ICV position (e.g., choke setting). For example, the ICV may be movable through several predefined positions (e.g., Position 1, Position 2, Position 3. . . Position n), where each position corresponds to a different amount of ICV opening. The magnitude of the change in the position of the ICV may thus be usable to determine, for example, if an ICV moved from Position 1 to Position 2, or from Position 1 to Position 3. The magnitude of the change in the position of the ICV may also be usable to determine if the ICV failed to reach an intended new position.

[0061] At block **508**, the result can be output and can be useable to control the ICV. For example, if it is determined based on the indicated magnitude of the change in the position of the ICV that the ICV failed to reach an intended new position, a computing system may cause an additional movement of the ICV to fully move the ICV to the new position. For example, when the computing device or a process control system communicatively coupled to the computing device includes an automated exception handling routine, a method for operating an ICV located downhole in a completed injection well may tune the ICV settings to achieve the target flow allocation per ICV within given tolerances. The exception handling routine may, for example, conditionally upon failure to move an ICV to a target position, move the ICV to a pre-defined value such as a closed position value and then move the ICV back to the target position. In another example, execution of

the exception handling routine may be triggered by a failure to detect movement of an ICV associated with a sent command to move the ICV despite the command for additional movement. In another example, execution of the exception handling routine may be triggered when the actual measured flow of the pressured fluid materially disagrees with the output of the trained machine-learning model for a given ICV position, or when the flow of the pressurized fluid deviates outside of an allowable range of target flow values. The action initiated by the exception handling routine may include specific activities tailored to the condition that triggered execution the exception handling routine. For example, routine events like flow rates above or below target flow rate tolerances may trigger an ICV shift to reduce or increase the ICV flow area and associated flow rate through the ICV, whereas a surface equipment failure may require a system re-set. In some examples, a failure to resolve an event that triggered execution of the exception handling routine may result in the generation of an alarm to one or more of the computing device, the process control system, or an operator.

[0062] According to aspects of the present disclosure, a system, a method, and a non-transitory computer-readable medium, are provided according to one or more of the following examples. As used below, any reference to a series of examples is to be understood as a reference to each of those examples disjunctively (e.g., “Examples 1-4” is to be understood as “Examples 1, 2, 3, or 4”).

[0063] Example 1 is a system comprising: a plurality of intelligent completion valves positionable downhole in a wellbore in a formation; a distributed acoustic sensing system positionable in the wellbore; a processor; and a memory communicatively coupled to the processor, the memory including instructions that are executable by the processor to cause the processor to perform operations comprising: storing a trained machine-learning model using training data comprising acoustic signal data generated by a pressurized injection fluid flowing outward through the plurality of intelligent completion valves at various flow rates and at different intelligent completion valve positions; receiving, from the distributed acoustic sensing system and in response to an amplitude spike in an acoustic signal associated with an intelligent completion valve of the plurality of intelligent completion valves, sensor data associated with the intelligent completion valve; applying the sensor data to the trained machine-learning model to generate a result determining that the amplitude spike is attributable to a change in the position of the intelligent completion valve; and outputting the result that is useable to control the intelligent completion valve.

[0064] Example 2 is the system of example 1, wherein the distributed acoustic sensing system includes an optical fiber cable located in the wellbore.

[0065] Example 3 is the system of example 1, wherein the change in position of the intelligent completion valve is a change in a choke setting of the intelligent completion valve.

[0066] Example 4 is the system of example 1, wherein the training data further comprises acoustic signal data generated by the pressurized injection fluid flowing outward through the plurality of intelligent completion valves at different pressurized injection fluid pressures.

[0067] Example 5 is the system of example 1, wherein the training data is buildable by: operating an injection logging tool in the wellbore to record flow rates of the pressurized injection fluid at locations near each of the intelligent completion valves; concurrently with operating the injection logging tool, operating the distributed acoustic sensing system to record acoustic signals generated at each of the intelligent completion valves; and while operating the injection logging tool and the distributed acoustic sensing system, controlling the intelligent completion valves such that: the position of the one of the plurality of intelligent completion valves through which the pressurized injection fluid flows is changed in a step-wise fashion.

[0068] Example 6 is the system of example 1, wherein training the machine-learning model comprises training the machine-learning model by supervised learning using a labeled dataset including: the acoustic signal data; pressurized injection fluid flow rate data; and engineered features selected from the group consisting of moveout-corrected stack semblance, ratio of short

time average to long time average, ratio of a root-mean-square amplitude of the acoustic signal before an amplitude spike to a root-mean-square amplitude of the acoustic signal after an amplitude spike, and combinations thereof.

[0069] Example 7 is the system of example 1, wherein in the acoustic signal data of the training data: a general increase in the amplitude of the acoustic signal over time following the amplitude spike, indicates an intelligent completion valve opening event; and a general decrease in the amplitude of acoustic signal over time following the amplitude spike indicates an intelligent completion valve closing event; wherein the operations further comprise, in response to detecting the general increase or the general decrease in the amplitude of the acoustic signal over time: determining that a rate of change in the amplitude of the acoustic signal has become stable at a time within a predefined length of time following the amplitude spike in the acoustic signal; and in response to determining that the amplitude of the acoustic signal has become stable within the predefined length of time, assigning the flow rate of the pressurized injection fluid at the time the rate of change in the amplitude of the acoustic signal became stable as the flow rate of the pressurized injection fluid through the intelligent completion valve at a current position of the intelligent completion valve.

[0070] Example 8 is a method, comprising: receiving, by a processor of a computing device, from a distributed acoustic sensing system positionable in a wellbore in a formation, sensor data associated with an intelligent completion valve of a plurality of intelligent completion valves positionable in the wellbore, the sensor data generated in response to an amplitude spike in an acoustic signal associated with the intelligent completion valve; applying the sensor data to a trained machine-learning model using training data comprising acoustic signal data generated by a pressurized injection fluid flowing outward through the plurality of intelligent completion valves at various flow rates and at different intelligent completion valve positions; generating a result determining that the amplitude spike is attributable to a change in the position of the intelligent completion valve; and outputting the result that is used to control the intelligent completion valve.

[0071] Example 9 is the method of example 8, wherein: the distributed acoustic sensing system includes an optical fiber cable located in the wellbore; and the change in position of the intelligent completion valve is a change in a choke setting of the intelligent completion valve.

[0072] Example 10 is the method of example 8, wherein the training data further comprises acoustic signal data generated by the pressurized injection fluid flowing outward through the plurality of intelligent completion valves at different pressurized injection fluid pressures.

[0073] Example 11 is the method of example 8, wherein the training data is built by: operating an injection logging tool in the wellbore to record flow rates of the pressurized injection fluid at locations near each of the intelligent completion valves; concurrently with operating the injection logging tool, operating the distributed acoustic sensing system to record acoustic signals generated at each of the intelligent completion valves; and while operating the injection logging tool and the distributed acoustic sensing system, controlling the intelligent completion valves such that: the position of the one of the plurality of intelligent completion valves through which the pressurized injection fluid flows is changed in a step-wise fashion.

[0074] Example 12 is the method of example 8, wherein the machine-learning model is trained by supervised learning using a labeled dataset including: the acoustic signal data; pressurized injection fluid flow rate data; and engineered features selected from the group consisting of moveout-corrected stack semblance, ratio of short time average to long time average, ratio of a root-mean-square amplitude of the acoustic signal before an amplitude spike to a root-mean-square amplitude of the acoustic signal after an amplitude spike, and combinations thereof.

[0075] Example 13 is the method of example 8, wherein in the acoustic signal data of the training data: a general increase in the amplitude of the acoustic signal over time following the amplitude spike, indicates an intelligent completion valve opening event; a general decrease in the amplitude of acoustic signal over time following the amplitude spike indicates an intelligent completion valve

closing event; and further comprising, in response to detecting the general increase or the general decrease in the amplitude of the acoustic signal over time: determining that a rate of change in the amplitude of the acoustic signal has become stable at a time within a predefined length of time following the amplitude spike in the acoustic signal; and in response to determining that the amplitude of the acoustic signal has become stable within the predefined length of time, assigning the flow rate of the pressurized injection fluid at the time the rate of change in the amplitude of the acoustic signal became stable as the flow rate of the pressurized injection fluid through the intelligent completion valve at a current position of the intelligent completion valve.

[0076] Example 14 is the method of example 8, further comprising determining a magnitude of the change in the position of the intelligent completion valve.

[0077] Example 15 is a non-transitory computer-readable medium comprising instructions that are executable by a processor of a computing device, for causing the processor to perform operations comprising: receiving, by a processor of a computing device, from a distributed acoustic sensing system positionable in a wellbore in a formation, sensor data associated with an intelligent completion valve of a plurality of intelligent completion valves positionable in the wellbore, the sensor data generated in response to an amplitude spike in an acoustic signal associated with the intelligent completion valve; applying the sensor data to a trained machine-learning model using training data comprising acoustic signal data generated by a pressurized injection fluid flowing outward through the plurality of intelligent completion valves at various flow rates and at different intelligent completion valve positions; generating a result determining that the amplitude spike is attributable to a change in the position of the intelligent completion valve; and outputting the result that is useable to control the intelligent completion valve.

[0078] Example 16 is the non-transitory computer-readable medium of example 15, wherein: the distributed acoustic sensing system includes an optical fiber cable located in the wellbore; and the change in position of the intelligent completion valve is a change in a choke setting of the intelligent completion valve.

[0079] Example 17 is the non-transitory computer-readable medium of example 15, wherein the training data further comprises acoustic signal data generated by the pressurized injection fluid flowing outward through the plurality of intelligent completion valves at different pressurized injection fluid pressures.

[0080] Example 18 is the non-transitory computer-readable medium of example 15, wherein the training data is buildable by: operating an injection logging tool in the wellbore to record flow rates of the pressurized injection fluid at locations near each of the intelligent completion valves; concurrently with operating the injection logging tool, operating the distributed acoustic sensing system to record acoustic signals generated at each of the intelligent completion valves; and while operating the injection logging tool and the distributed acoustic sensing system, controlling the intelligent completion valves such that: the position of the one of the plurality of intelligent completion valves through which the pressurized injection fluid flows is changed in a step-wise fashion.

[0081] Example 19 is the non-transitory computer-readable medium of example 15, wherein training the machine-learning model comprises training the machine-learning model by supervised learning using a labeled dataset including: the acoustic signal data; pressurized injection fluid flow rate data; and engineered features selected from the group consisting of moveout-corrected stack semblance, ratio of short time average to long time average, ratio of a root-mean-square amplitude of the acoustic signal before an amplitude spike to a root-mean-square amplitude of the acoustic signal after an amplitude spike, and combinations thereof.

[0082] Example 20 is the non-transitory computer-readable medium of example 15, wherein in the acoustic signal data of the training data: a general increase in the amplitude of the acoustic signal over time following the amplitude spike indicates an intelligent completion valve opening event; and a general decrease in the amplitude of acoustic signal over time following the amplitude spike

indicates an intelligent completion valve closing event; wherein the operations further comprise, in response to detecting the general increase or the general decrease in the amplitude of the acoustic signal over time: determining that a rate of change in the amplitude of the acoustic signal has become stable at a time within a predefined length of time following the amplitude spike in the acoustic signal; and in response to determining that the amplitude of the acoustic signal has become stable within the predefined length of time, assigning the flow rate of the pressurized injection fluid at the time the rate of change in the amplitude of the acoustic signal became stable as the flow rate of the pressurized injection fluid through the intelligent completion valve at a current position of the intelligent completion valve.

[0083] The foregoing description of certain examples, including illustrated examples, has been presented only for the purpose of illustration and description and is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Numerous modifications, adaptations, and uses thereof will be apparent to those skilled in the art without departing from the scope of the disclosure.

Claims

1. A system comprising: a plurality of intelligent completion valves positionable downhole in a wellbore in a formation; a distributed acoustic sensing system positionable in the wellbore; a processor; and a memory communicatively coupled to the processor, the memory including instructions that are executable by the processor to cause the processor to perform operations comprising: storing a trained machine-learning model using training data comprising acoustic signal data generated by a pressurized injection fluid flowing outward through the plurality of intelligent completion valves at various flow rates and at different intelligent completion valve positions; receiving, from the distributed acoustic sensing system and in response to an amplitude spike in an acoustic signal associated with an intelligent completion valve of the plurality of intelligent completion valves, sensor data associated with the intelligent completion valve; applying the sensor data to the trained machine-learning model to generate a result determining that the amplitude spike is attributable to a change in the position of the intelligent completion valve; and outputting the result that is useable to control the intelligent completion valve.
2. The system of claim 1, wherein the distributed acoustic sensing system includes an optical fiber cable located in the wellbore.
3. The system of claim 1, wherein the change in position of the intelligent completion valve is a change in a choke setting of the intelligent completion valve.
4. The system of claim 1, wherein the training data further comprises acoustic signal data generated by the pressurized injection fluid flowing outward through the plurality of intelligent completion valves at different pressurized injection fluid pressures.
5. The system of claim 1, wherein the training data is buildable by: operating an injection logging tool in the wellbore to record flow rates of the pressurized injection fluid at locations near each of the intelligent completion valves; concurrently with operating the injection logging tool, operating the distributed acoustic sensing system to record acoustic signals generated at each of the intelligent completion valves; and while operating the injection logging tool and the distributed acoustic sensing system, controlling the intelligent completion valves such that: the position of the one of the plurality of intelligent completion valves through which the pressurized injection fluid flows is changed in a step-wise fashion.
6. The system of claim 1, wherein training the machine-learning model comprises training the machine-learning model by supervised learning using a labeled dataset including: the acoustic signal data; pressurized injection fluid flow rate data; and engineered features selected from the group consisting of moveout-corrected stack semblance, ratio of short time average to long time average, ratio of a root-mean-square amplitude of the acoustic signal before an amplitude spike to a

root-mean-square amplitude of the acoustic signal after an amplitude spike, and combinations thereof.

7. The system of claim 1, wherein in the acoustic signal data of the training data: a general increase in the amplitude of the acoustic signal over time following the amplitude spike, indicates an intelligent completion valve opening event; and a general decrease in the amplitude of acoustic signal over time following the amplitude spike indicates an intelligent completion valve closing event; wherein the operations further comprise, in response to detecting the general increase or the general decrease in the amplitude of the acoustic signal over time: determining that a rate of change in the amplitude of the acoustic signal has become stable at a time within a predefined length of time following the amplitude spike in the acoustic signal; and in response to determining that the amplitude of the acoustic signal has become stable within the predefined length of time, assigning the flow rate of the pressurized injection fluid at the time the rate of change in the amplitude of the acoustic signal became stable as the flow rate of the pressurized injection fluid through the intelligent completion valve at a current position of the intelligent completion valve.

8. A method, comprising: receiving, by a processor of a computing device, from a distributed acoustic sensing system positionable in a wellbore in a formation, sensor data associated with an intelligent completion valve of a plurality of intelligent completion valves positionable in the wellbore, the sensor data generated in response to an amplitude spike in an acoustic signal associated with the intelligent completion valve; applying the sensor data to a trained machine-learning model using training data comprising acoustic signal data generated by a pressurized injection fluid flowing outward through the plurality of intelligent completion valves at various flow rates and at different intelligent completion valve positions; generating a result determining that the amplitude spike is attributable to a change in the position of the intelligent completion valve; and outputting the result that is used to control the intelligent completion valve.

9. The method of claim 8, wherein: the distributed acoustic sensing system includes an optical fiber cable located in the wellbore; and the change in position of the intelligent completion valve is a change in a choke setting of the intelligent completion valve.

10. The method of claim 8, wherein the training data further comprises acoustic signal data generated by the pressurized injection fluid flowing outward through the plurality of intelligent completion valves at different pressurized injection fluid pressures.

11. The method of claim 8, wherein the training data is built by: operating an injection logging tool in the wellbore to record flow rates of the pressurized injection fluid at locations near each of the intelligent completion valves; concurrently with operating the injection logging tool, operating the distributed acoustic sensing system to record acoustic signals generated at each of the intelligent completion valves; and while operating the injection logging tool and the distributed acoustic sensing system, controlling the intelligent completion valves such that: the position of the one of the plurality of intelligent completion valves through which the pressurized injection fluid flows is changed in a step-wise fashion.

12. The method of claim 8, wherein the machine-learning model is trained by supervised learning using a labeled dataset including: the acoustic signal data; pressurized injection fluid flow rate data; and engineered features selected from the group consisting of moveout-corrected stack semblance, ratio of short time average to long time average, ratio of a root-mean-square amplitude of the acoustic signal before an amplitude spike to a root-mean-square amplitude of the acoustic signal after an amplitude spike, and combinations thereof.

13. The method of claim 8, wherein in the acoustic signal data of the training data: a general increase in the amplitude of the acoustic signal over time following the amplitude spike, indicates an intelligent completion valve opening event; a general decrease in the amplitude of acoustic signal over time following the amplitude spike indicates an intelligent completion valve closing event; and further comprising, in response to detecting the general increase or the general decrease in the amplitude of the acoustic signal over time: determining that a rate of change in the amplitude

of the acoustic signal has become stable at a time within a predefined length of time following the amplitude spike in the acoustic signal; and in response to determining that the amplitude of the acoustic signal has become stable within the predefined length of time, assigning the flow rate of the pressurized injection fluid at the time the rate of change in the amplitude of the acoustic signal became stable as the flow rate of the pressurized injection fluid through the intelligent completion valve at a current position of the intelligent completion valve.

14. The method of claim 8, further comprising determining a magnitude of the change in the position of the intelligent completion valve.

15. A non-transitory computer-readable medium comprising instructions that are executable by a processor of a computing device, for causing the processor to perform operations comprising: receiving, by a processor of a computing device, from a distributed acoustic sensing system positionable in a wellbore in a formation, sensor data associated with an intelligent completion valve of a plurality of intelligent completion valves positionable in the wellbore, the sensor data generated in response to an amplitude spike in an acoustic signal associated with the intelligent completion valve; applying the sensor data to a trained machine-learning model using training data comprising acoustic signal data generated by a pressurized injection fluid flowing outward through the plurality of intelligent completion valves at various flow rates and at different intelligent completion valve positions; generating a result determining that the amplitude spike is attributable to a change in the position of the intelligent completion valve; and outputting the result that is useable to control the intelligent completion valve.

16. The non-transitory computer-readable medium of claim 15, wherein: the distributed acoustic sensing system includes an optical fiber cable located in the wellbore; and the change in position of the intelligent completion valve is a change in a choke setting of the intelligent completion valve.

17. The non-transitory computer-readable medium of claim 15, wherein the training data further comprises acoustic signal data generated by the pressurized injection fluid flowing outward through the plurality of intelligent completion valves at different pressurized injection fluid pressures.

18. The non-transitory computer-readable medium of claim 15, wherein the training data is buildable by: operating an injection logging tool in the wellbore to record flow rates of the pressurized injection fluid at locations near each of the intelligent completion valves; concurrently with operating the injection logging tool, operating the distributed acoustic sensing system to record acoustic signals generated at each of the intelligent completion valves; and while operating the injection logging tool and the distributed acoustic sensing system, controlling the intelligent completion valves such that: the position of the one of the plurality of intelligent completion valves through which the pressurized injection fluid flows is changed in a step-wise fashion.

19. The non-transitory computer-readable medium of claim 15, wherein training the machine-learning model comprises training the machine-learning model by supervised learning using a labeled dataset including: the acoustic signal data; pressurized injection fluid flow rate data; and engineered features selected from the group consisting of moveout-corrected stack semblance, ratio of short time average to long time average, ratio of a root-mean-square amplitude of the acoustic signal before an amplitude spike to a root-mean-square amplitude of the acoustic signal after an amplitude spike, and combinations thereof.

20. The non-transitory computer-readable medium of claim 15, wherein in the acoustic signal data of the training data: a general increase in the amplitude of the acoustic signal over time following the amplitude spike indicates an intelligent completion valve opening event; and a general decrease in the amplitude of acoustic signal over time following the amplitude spike indicates an intelligent completion valve closing event; wherein the operations further comprise, in response to detecting the general increase or the general decrease in the amplitude of the acoustic signal over time: determining that a rate of change in the amplitude of the acoustic signal has become stable at a time within a predefined length of time following the amplitude spike in the acoustic signal; and in response to determining that the amplitude of the acoustic signal has become stable within the

predefined length of time, assigning the flow rate of the pressurized injection fluid at the time the rate of change in the amplitude of the acoustic signal became stable as the flow rate of the pressurized injection fluid through the intelligent completion valve at a current position of the intelligent completion valve.
